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An interpretable machine learning framework for explaining company valuation

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ABSTRACT

Valuing private enterprises — particularly early-stage firms — remains a significant challenge due to high variability and frequent overvaluation, often exceeding 50%. These inaccuracies undermine trust in the venture capital (VC) market and complicate investment decision-making. To address these issues, we present machine learning (ML) frameworks that enhance valuation accuracy. Our study introduces a novel ML approach that outperforms traditional models, achieving a Mean Absolute Percentage Error of 34.90%, significantly improving upon industry benchmarks. The Total Cost of Ownership is identified as the most critical valuation methodology, with financial metrics becoming increasingly influential as firm value rises. Our findings highlight the potential of advanced data mining techniques to deliver more reliable, comprehensive valuations within the VC landscape. Future research should explore expanding this analysis to emerging markets and incorporating primary data for deeper insights.

1. Introduction

Valuing companies, especially in technology-driven sectors, is a cornerstone of innovation management and entrepreneurship. Accurate valuations directly impact strategic decisions regarding mergers, acquisitions, investment allocations, and the commercialisation of technological innovations. However, the valuation process is inherently complex due to the diverse and often intangible factors involved, particularly when assessing early-stage firms in the venture capital (VC) market. These firms, typically characterised by high volatility, lack comprehensive historical financial data and often over-rely on intellectual capital, leading to frequent overvaluations that can exceed 50% [1]. This problem not only undermines investor confidence but also inhibits effective capital allocation in markets driven by technological advancements.

Despite the pivotal role that precise valuations play in supporting VC investments and technological innovation, a significant gap persists in the integration of advanced, data-driven methodologies into the valuation process. Traditional approaches, which predominantly focus on financial metrics, fail to capture the full scope of value drivers for startups, such as technological potential, scalability, and intellectual capital [2]. As the management of technological innovation increasingly depends on understanding both tangible and intangible assets, there is a pressing need for more holistic and adaptive valuation frameworks.

In response to this challenge, our research proposes a novel machine learning (ML)-based valuation framework tailored to the specific needs of technology-intensive startups. By leveraging multiple valuation methodologies and incorporating company characteristics into an ML-driven model, we aim to improve valuation accuracy and offer more reliable tools for investors, entrepreneurs, and policymakers. This approach aligns with recent trends in technological innovation management, where advanced data analytics and artificial intelligence (AI) are reshaping traditional business processes [3]. Moreover, our framework reflects the shift towards data-driven decision-making in entrepreneurship, particularly in managing high-growth, high-risk enterprises [4].

The complexity of startup valuation, particularly in the context of innovation-driven firms, arises from the myriad of factors that must be evaluated, including market potential, intellectual property, and technological advancements. Misvaluations not only lead to inflated valuations but also impair the ability of startups to attract further investment, ultimately stifling innovation and economic growth [5].

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Therefore, developing a framework that addresses these issues is critical for ensuring the sustainable growth of tech-based enterprises and facilitating the commercialisation of technological innovations.

This paper contributes to the literature by addressing key gaps in the intersection of entrepreneurship, technological innovation, and valuation practices. Specifically, we focus on how advanced ML techniques can be applied to enhance the accuracy of valuations in the VC landscape. We explore the effects of different valuation methodologies on firm valuation and investigate how company characteristics influence valuation outcomes at various stages of development. Through this lens, our study seeks to answer the following research questions: (1) Which valuation methodologies are most effective for technology-driven startups? (2) How do company characteristics, such as innovation potential and intellectual capital, impact valuation accuracy? (3) What are the broader implications for technological innovation management and entrepreneurial investment strategies?

The core contribution of this research lies in its development of four ML frameworks, including a novel approach that outperforms traditional neural network (NN) models. These models are validated against industry benchmarks, utilising explainability techniques such as Local Interpretable Model-Agnostic Explanations (LIME) to provide insights into the key factors influencing valuation accuracy. This analysis advances the theoretical understanding of the valuation process, particularly in the context of innovation-intensive enterprises, and offers practical tools for investors and entrepreneurs navigating the uncertainties of the VC market.

The remainder of this paper is structured as follows. In Section 2, we review the current state of ML applications in finance, with an emphasis on firm valuation and the gaps in existing methodologies. Section 3 outlines the proposed ML frameworks and the data mining (DM) techniques employed. Section 4 presents the results of our analysis, including the validation of our models and a discussion on the theoretical and practical implications for the fields of entrepreneurship and innovation management. Finally, Section 5 summarises the key findings and proposes future research directions, particularly in expanding the application of these techniques to emerging markets and technology-based firms.

2. Literature review

This section provides a comprehensive review of the application of ML techniques in finance, with a focus on firm valuation, entrepreneurial finance, and technological innovation. The review is organised into three subsections: significant developments in ML within the financial industry (Section 2.1), specific applications of ML in firm valuation (Section 2.2), and the identification of key knowledge gaps that this study seeks to address (Section 2.3).

2.1. Machine learning in the financial industry

The financial industry has seen significant advancements in the application of ML techniques, particularly in areas such as risk assessment, investment strategies, and financial forecasting [6]. These advancements have demonstrated the versatility of ML across a wide range of financial tasks, including bankruptcy prediction [7,8], credit rating assessments, and stock market trend detection [9,10]. The literature on financial time series forecasting has been particularly influential, with studies such as [11] demonstrating the potential of ML to predict market movements with high accuracy — an essential capability in a field where precision can directly impact economic outcomes.

A wide range of empirical studies have validated the effectiveness of ML in financial applications. For instance, [12] applied support vector machines (SVM) to financial risk prediction, while [13] utilised NNs to identify companies in financial distress. Such models have been used not only for predictive tasks but also for automating complex decision-making processes, as seen in the development of automated trading systems [14]. These examples highlight how ML has been integrated into financial systems to enhance decision-making capabilities.

In the context of entrepreneurship and VC, ML is gaining attention for its potential to predict startup success and identify high-growth firms [15]. For example, [16] applied ML to predict the survival of new ventures, while [17] used ML to assess entrepreneurial skills. The adoption of ML in entrepreneurial finance represents an exciting frontier in both fields, suggesting that ML can provide deeper insights into the complex dynamics of startup ecosystems.

As technological innovations continue to reshape industries, the integration of ML into entrepreneurial finance offers an opportunity to enhance decision-making in the VC landscape. Recent interdisciplinary research highlights the need for collaborative approaches that combine data science, entrepreneurship, and finance to address emerging challenges in the innovation ecosystem [4,18]. These developments underscore the growing relevance of ML as a transformative tool in the financial and entrepreneurial sectors. Table 1 provides a comprehensive overview of ML applications in firm valuation as documented in the literature.

2.2. Applications of machine learning to firm valuation

While ML has shown considerable promise across financial applications, its use in firm valuation, particularly in the context of startups and innovation-driven firms, remains relatively underexplored. Firm valuation presents unique challenges due to the high degree of variability in early-stage companies, the lack of historical financial data, and the prominence of intangible assets such as intellectual capital and technological potential [32,33]. These complexities make traditional valuation approaches inadequate for assessing the full range of value drivers in technology-driven enterprises.

However, some studies have successfully demonstrated the potential of ML in firm valuation. [28], for example, used predictive modelling to forecast funding events, while [21] applied ML to estimate startup valuations where financial data was sparse or unavailable. Similarly, [19] explored ML for investment performance analysis, highlighting the role of advanced data analytics in improving the accuracy of investment decisions in the VC industry.

Moreover, ML-based frameworks have been used to analyse both financial and non-financial factors, providing a more comprehensive approach to firm valuation. [24] studied the growth of Finnish firms using ML to evaluate a combination of financial and non-financial variables, while [27] introduced a rule-based expert system for predicting company acquisitions. These studies demonstrate the versatility of ML in handling the complex, multi-dimensional nature of firm valuation.

The integration of ML into firm valuation is further supported by recent advancements in combining supervised and unsupervised learning techniques. For instance, Haddad et al. [22] used a hybrid approach that combined SVM with clustering to identify business models with high growth potential. [29] employed big data and ML to remove psychological bias from asset pricing, offering a novel approach to firm valuation based on observable asset prices and panel data. Additionally, decision aids that combine both cognitive and strategic criteria have been shown to improve the selection accuracy of startups, providing valuable perspectives when traditional financial metrics fail to offer sufficient insight [34]. This aligns with the growing recognition of the importance of non-financial factors, such as intellectual capital, which often play a crucial role in early-stage venture success. These methods underscore the potential of ML to provide more accurate and scalable solutions to firm valuation challenges in technology-driven industries.

Despite these advancements, the use of ML in firm valuation remains a relatively nascent field, particularly in its application to early-stage, innovation-intensive firms. The ability to capture the full scope of a startup's potential — its intellectual property, technological innovation, and market adaptability — remains a critical area for further exploration.

Table 1
Overview of ML applications in firm valuation.

Reference	Purpose	ML technique(s)	Key findings
[19]	Predicting new venture survival	Gradient Boosted Decision Trees	ML algorithms consistently generate higher investment returns than angel investors on angel investment platforms.
[20]	Predicting future pairs of the acquirer and acquiree companies	SHapley Additive exPlanations (SHAP) value interpretation	Newly considered network and company relation features demonstrate significant importance, with 10 out of the top 22 highly influential features.
[21]	Inferring startup valuation corresponding to announced funding rounds	Supervised, unsupervised and semi-supervised domain adaptation	The proposed method outperforms previous approaches in the literature by a significant margin. The Domain Adaptation framework proves suitable for the task.
[22]	Predicting business models with growth potential and survival prospects	SVM and clustering	Some business models exhibit strong associations with different variables affecting startup success.
[23]	Predicting startup success	Trees, Bayesian and Rules-based algorithms	Start-up success is predicted with 83.76% accuracy.
[24]	Examining factors influencing high growth in new firms	Logistic regression analysis	Clear connection between entrepreneurs' know-how and firms' high growth. External networks also accelerate growth.
[25]	Predicting winning and non-winning companies	LASSO, SVM, Boosted Regression	Modern ML methods do not significantly improve predictions, indicating the challenge of selecting competition winners.
[26]	Examining the relationship between startup valuation and funding amount	Linear (log–log space) model and Bayesian model	Statistically significant results provide practitioners with a tool to estimate post-money valuations
[27]	Classifying firms as acquisition candidates based on financial and other data	IXL, a rule induction technique	A rule induction technique aids financial analysts and investors in buy/sell decisions
[28]	Predicting early-stage startup success	Gradient boosting (GB)	Utilising companies' web mentions significantly enhances predictive performance compared to using structured data alone.
[29]	Separating intrinsic equity value from mispricing caused by irrational investor sentiments	Drift Detection Method, SVM and Recurrent Inference Machines	The approach confirms its validity in separating intrinsic value from sentiment-driven mispricing.
[30]	Pricing and benchmarking Enterprise Value for valuation and credit risk assessment	Decision Trees (DT), Bagging Trees, SVM, Gaussian Process Regression	Features related to value creation are crucial while incorporating industry/country-specific variables improves ML performance.
[31]	Predicting the next outcome of a company: acquisition, closure, initial public offering, funding round, or no event	SVM, DT, Random Forests (RF), Extremely Randomised Trees, and GB.	ML aids venture investors in decision-making, uncovering opportunities and assessing investment risks more effectively.

2.3. Knowledge gap

While ML has made significant inroads into the financial industry, gaps remain in understanding how these techniques can be fully leveraged for firm valuation, especially in high-growth, technology-driven sectors. The complexity of startup ecosystems, where financial and non-financial value drivers interplay, presents unique challenges that traditional models cannot address. Existing studies have often been constrained by limited datasets or have focused narrowly on financial variables, which do not capture the dynamic nature of technology-based firms [31].

This study seeks to address these gaps by developing a robust, scalable ML framework that incorporates both financial and non-financial factors into the valuation process. By leveraging a comprehensive dataset of Nordic startups, the research aims to build predictive models that account for the unique characteristics of technology-driven firms, including their intellectual capital, market potential, and scalability. This approach not only enhances valuation accuracy but also provides deeper insights into the key determinants of firm value throughout different stages of company growth.

Additionally, the study emphasises the importance of non-financial valuation methodologies, particularly those that assess intangible assets such as innovation potential and intellectual property. These methods complement traditional accounting-based approaches and are essential for accurately valuing early-stage, technology-intensive firms. In this regard, the research contributes to the growing body of work that seeks to integrate non-traditional value drivers into the broader discourse on firm valuation.

Finally, while secondary data sources offer valuable insights, this study acknowledges their limitations and calls for future research that incorporates primary data, such as survey responses from entrepreneurs and investors. Such data would offer a more nuanced understanding of the valuation process, particularly in emerging markets, where startup ecosystems are rapidly evolving but remain under-researched.

In summary, this literature review highlights the progress that has been made in applying ML to finance and firm valuation, while also identifying critical areas where further research is needed. By developing an ML framework tailored to the needs of technology-intensive startups, this study aims to contribute to the broader fields of technological innovation and entrepreneurship, providing actionable insights for both practitioners and researchers.

3. Methodology

This section outlines the methodology employed in this study, focusing on two key components: the DM process and the ML approach to company valuation. The methodology is designed to ensure a robust and adaptable framework for analysing complex datasets and developing accurate company valuations, with an emphasis on technological innovation and its application in VC decision-making.

The methodology follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology, a widely adopted framework in data science, tailored to the specific requirements of the VC industry and the startup ecosystem (Section 3.1). It also describes the construction of the valuation models and the datasets employed (Section 3.2), with an emphasis on integrating ML techniques that closely resemble real-world VC decision-making processes.

3.1. Machine learning process

The ML process in this study follows the CRISP-DM methodology, dividing the process into six distinct phases that ensure a structured and repeatable approach. This framework allows for systematic data exploration, preparation, and model development while addressing the complexities of firm valuation in innovation-driven industries. The CRISP-DM phases are tailored to meet the needs of the VC context, where firm valuation is influenced by a multitude of both financial and non-financial factors. Below is an overview of the six phases:

1. *Business understanding.* Section 1 establishes the context of this research, presenting the business and scientific objectives within the VC landscape. The emphasis is on developing a comprehensive valuation model that integrates multiple methodologies and leverages ML to address the unique challenges faced by early-stage, innovation-intensive firms.
2. *Data understanding.* Section 3.1.1 provides an in-depth overview of the dataset, focusing on company characteristics and the valuation methods used. The diversity of these characteristics highlights the complexities involved in accurately valuing startups and the need for a comprehensive data-driven approach.
3. *Data preparation.* Data preprocessing is critical in ensuring that the models are built on reliable data. In Section 3.1.2, we discuss the handling of missing values, normalisation, and data augmentation, focusing on addressing challenges such as data sparsity and high dimensionality, which are common in VC datasets.
4. *Modelling.* Section 3.1.3 outlines the ML algorithms employed in this study. The algorithms are selected based on their suitability for predictive modelling in the context of firm valuation, with a focus on explainability, scalability, and real-world applicability.
5. *Model evaluation.* Evaluation metrics such as the coefficient of determination (R^2) and Mean Absolute Percentage Error (MAPE) are used to assess model performance. These metrics are essential for determining the model's predictive accuracy and for identifying key valuation drivers, as discussed in Section 3.1.4.
6. *Deployment.* Although this study does not focus on deployment, the potential real-world applications of the models are acknowledged, particularly in the context of VC decision-making, where accurate and interpretable firm valuations can significantly influence investment outcomes.

3.1.1. Data understanding

This subsection describes the data collection process, focusing on company characteristics and intermediate valuations. The primary dataset consists of company characteristics obtained from a proprietary operational database. To ensure data integrity and scalability, a data warehouse and data staging area were developed, providing a secure and up-to-date environment for data analysis. The architecture of the data processing system showcases the flow of information from the operational database to the final analytical datasets.

A star schema was employed to organise the data into a central fact table surrounded by dimension tables, each representing different company characteristics. This design allows for efficient querying and analysis, particularly when dealing with large and complex datasets. The process of transforming JSON files into a snowflake model further enhances the ability to work with diverse data sources.

The company characteristics dataset consists of 73 observations and 1308 variables, covering a wide range of financial and non-financial factors critical to firm valuation. To maintain confidentiality, the analysis focuses on aggregated data, ensuring that proprietary information remains protected. The literature review identified key company characteristics relevant to firm valuation, summarised in Table 2. These characteristics serve as the foundation for the intermediate valuations derived in this study.

The set of intermediate valuation methodologies selected for this study reflects a broad spectrum of financial and non-financial perspectives on firm value. These methods were chosen based on three criteria: (i) established use in VC or private equity valuation practice, (ii) capacity to represent both tangible and intangible value drivers, and (iii) availability of underlying data.

For instance, Discounted Cash Flows (DCF) is widely regarded as a theoretically grounded benchmark in corporate finance, but requires detailed financial forecasts that are often unavailable or unreliable for early-stage startups. Economic Value Added (EVA) and Cash Value Added (CVA) offer performance-based alternatives that incorporate the cost of capital but depend on robust estimates of capital employed and discount rates. Value Added Intellectual Coefficient (VAIC) provides a framework for assessing intellectual capital efficiency, which is particularly relevant for knowledge-intensive startups. Total Cost of Ownership (TCO), although more commonly applied in information technology and infrastructure procurement, captures holistic cost-related information that complements other valuation perspectives and shows the highest predictive importance in our models. The Checklist Method and Balanced Scorecard (BSC) allow for semi-structured, expert-driven assessments when quantitative data is limited. Graham's Intrinsic Value (IV) and Book Value, while more traditional and accounting-focused, serve as useful baselines. This pluralistic approach ensures that the valuation ensemble captures the multidimensional nature of startup value while maintaining applicability across heterogeneous data conditions.

Book Value. The BV valuation methodology is a financial analysis technique that determines the value of a company based on its net assets.

$$Book\ Value = Total\ Assets - Total\ Liabilities$$

where *Total Assets* represents the sum of all assets owned by the company, including tangible and intangible assets, and *Total Liabilities* is the sum of all debts and obligations owed by the company.

Graham's Intrinsic Value. Graham's IV methodology is a fundamental analysis technique that estimates a company's true worth based on expected earnings and growth. A revised version of the formula, adapted for modern contexts, incorporates a yield adjustment to account for interest rate changes:

$$Intrinsic\ Value\ per\ Share = \frac{EPS \times (8.5 + 2 \times g) \times 4.4}{Y}$$

where EPS is the earnings per share, g is the expected annual earnings growth rate, and Y is the current yield on AAA-rated corporate bonds. This approach adjusts the original Graham formula to reflect the opportunity cost of capital.

To estimate the value of the whole company:

$$Company\ IV = Intrinsic\ Value\ per\ Share \times Shares\ Outstanding$$

Discounted Cash Flows. To estimate the fair value of an investment using the DCF method, the enterprise value is derived by projecting future cash flows or earnings and calculating the terminal value. This enterprise value is then discounted to the present value using the WACC, which reflects the company's debt and equity costs. The valuer adjusts the enterprise value by considering surplus assets, non-operating assets, and excess liabilities. Additionally, they account for financial instruments that rank higher than the firm's instrument in a liquidation scenario and assess the impact of potential dilution on the investment. Finally, the attributable enterprise value is allocated among the relevant financial instruments based on market participant perspectives [44,45].

$$DCF\ Value = \frac{Cash\ Flow\ Year\ 1}{(1 + Discount\ Rate)^1} + \frac{Cash\ Flow\ Year\ 2}{(1 + Discount\ Rate)^2} + \cdots + \frac{Cash\ Flow\ Year\ N}{(1 + Discount\ Rate)^N}$$

Table 2
Summary of relevant company characteristics used in comparable literature.

Study	Scope	Relevant company characteristics
[35]	Firm valuation	Financial capital and human capital, industry and market, product and technology, attractive industries, high-quality founders and top management teams, strong external relationships
[36]	Firm valuation	Twitter sentiment
[37]	Firm valuation	Network, knowledge sharing
[38]	Firm valuation	Real interest rate, consumer price index, gross domestic product, weighted average cost of capital (WACC)
[39]	Firm valuation	Internal resources, industry resources, network-based resources
[21]	Firm valuation	Age, country, founders, employees, news, funding round, financial history, social network
[19]	Firm valuation	Online legitimacy, initial funding and legal form, market timing, founder experiences, industry, founder education
[28]	Firm valuation	Age, industry, websites, offices, description, products, investor-related factors, team-related factors
[40]	Investment decision-making	Human capital, financial capital, market timing, industry timing, online legitimacy
[41]	Investment decision-making	Marketability, market potential, industry features
[42]	Intellectual asset valuation	Research and development intensity, industry, advertising intensity, family, cash flow rights, cash flow participation, cash flow nonparticipation, business group, board independence, blockholder, sale growth, size, leverage, capital intensity, dividend, profitability, age, export, analyst following, concentration, market share
[40]	Investment decision-making	Revenue growth, product/service value-added, management team track record, profitability
[43]	Investment decision-making	Product adoption, status, protectability, customer engagement, route to market, market potential, relevant experience, financial model
[34]	Investment decision-making	Strategy, team and finance

where *Cash Flow Year i* refers to the expected cash flow in year *i* and *Discount Rate* is the rate used to discount the future cash flows — it represents the investor's required rate of return and considers the risk associated with the investment.

Terminal Growth. The TG methodology is another widely used approach for estimating a company's value. It applies a TG rate to account for the expected cash flows beyond the explicit forecast period. This methodology assumes that companies will continue to generate cash flows in the long term. The free cash flows beyond the explicit forecast period are projected to calculate the terminal value, and a TG rate is applied. Typically, the TG rate is based on the expected long-term growth rate of the company's industry. The resulting terminal value is then discounted to the present value using the WACC as the discount rate. Adding the present value of the explicit forecast period cash flows to the terminal value provides the company's total enterprise value.

$$Terminal\ Value = \frac{Cash\ Flow\ Year\ N \times (1 + Terminal\ Growth\ Rate)}{Discount\ Rate - Terminal\ Growth\ Rate}$$

where *Cash Flow Year N* refers to the expected cash flow in the last projected year, *Terminal Growth Rate* represents the assumed growth rate for cash flows beyond the projection period, and *Discount Rate* is the rate used to discount the terminal value back to its present value.

Benchmark. The benchmark valuation methodology compares a company to similar firms in the market. It involves identifying comparable companies, gathering their financial data, and applying their average or median multiples to the valued company. This approach is simple, market-based, and helpful for assessing relative valuation. However, it relies on accurate data, assumes market accuracy, and may oversimplify or overlook unique company characteristics. It serves as a reference point when used alongside other valuation methods.

$$Company\ Value = Benchmark\ Multiple \times Financial\ Metric$$

where *Benchmark Multiple* refers to the valuation multiple derived from similar companies or industry standards, and *Financial Metric* represents the relevant financial metric of the company being valued, such as earnings, sales, or EBITDA.

Economic Value Added. The EVA methodology is a financial analysis technique that measures the value created by a company for its shareholders. The rationale behind this methodology is that traditional accounting profits do not fully reflect the actual value created by a company, and therefore, a more comprehensive measure is needed. EVA is calculated by subtracting the cost of capital from the after-tax operating profit of a company.

$$EVA = Net\ Operating\ Profit\ After\ Tax\ (NOPAT) - Capital\ Charge$$

where *NOPAT* refers to the company's operating profit after deducting taxes but before deducting the cost of capital and *Capital Charge* represents the cost of the capital employed in the business — it is calculated by multiplying the invested capital by the company's cost of capital, often expressed as a percentage.

Cash Value Added. CVA is a financial performance metric similar to EVA, but it focuses on after-tax operating cash flows. It evaluates whether a company generates sufficient cash to cover the cost of capital:

$$CVA = Operating\ Cash\ Flow - (Invested\ Capital \times WACC)$$

where *Operating Cash Flow* is the cash generated from operations, *Invested Capital* is the total capital employed in the business, and *WACC* is the weighted average cost of capital.

Total Cost of Ownership. TCO is an accounting-based methodology used to estimate the comprehensive cost of acquiring and maintaining an asset. Although not a traditional valuation technique in finance, it complements decision-making by considering the life-cycle costs of ownership:

$$TCO = Acquisition\ Cost + Operating\ Costs + Maintenance\ Costs + Disposal\ Costs$$

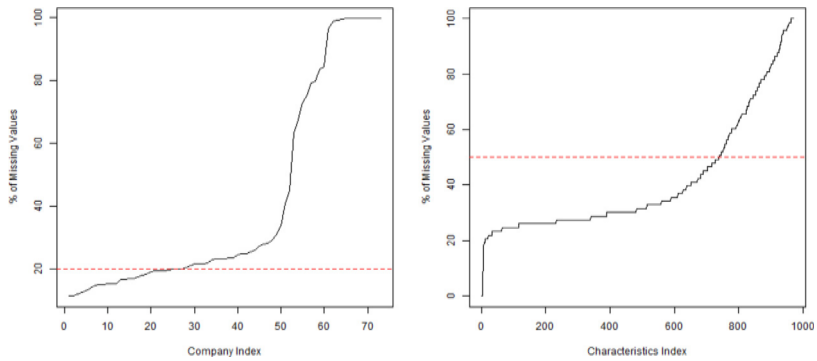

Fig. 1. Distribution of missing values as a percentage of each characteristic per company.

where the terms reflect the full cost over the asset's life. TCO is especially relevant in information technology, infrastructure, or capital equipment procurement, but may inform cost-side inputs in broader valuation assessments.

Balanced Scorecard. The BSC is a strategic management framework that translates an organisation's mission and vision into performance objectives across four perspectives: financial, customer, internal processes, and learning and growth. It is not a monetary valuation model, but can be used to assess non-financial drivers of value. While it does not yield a direct firm value estimate, composite performance can be summarised using weighted scoring:

$$Composite\ Score = \sum_{i=1}^{N} Weight_i \times Score_i$$

where Weight_i represents the strategic importance of each key performance indicator and Score_i its performance rating. These insights may support qualitative valuation adjustments.

Value Added Intellectual Coefficient. The VAIC measures the efficiency of intellectual capital utilisation. It is composed of three elements: human capital efficiency (HCE), structural capital efficiency (SCE), and capital employed efficiency (CEE):

$$VAIC = HCE + SCE + CEE$$

where: VA = Operating Profit+Depreciation+Amortisation, $\text{HCE} = \frac{\text{VA}}{\text{HC}}$, with HC as total salaries and wages, $\text{SCE} = \frac{\text{SC}}{\text{VA}}$, with SC = VA − HC, and $\text{CEE} = \frac{\text{VA}}{\text{CE}}$, with CE as capital employed (net assets). This model is primarily used for assessing knowledge-based firms and provides an indirect indication of firm performance related to intellectual assets.

Overview. The dataset includes 11 intermediate valuation methodologies, which provide a comprehensive view of firm value across different dimensions. The summary of these intermediate valuations is presented in Table 3.

3.1.2. Data preparation

The data preparation phase involved cleaning, transforming, and augmenting the dataset to address common challenges in firm valuation, such as missing values (see Fig. 1) and high dimensionality. Given the limited number of observations and the sparsity of the data, specific preprocessing steps were taken to ensure that the dataset was suitable for ML analysis. Fig. 2 presents a sensitivity matrix for the subject.

Missing values were addressed through a structured two-step procedure: threshold-based exclusion followed by median imputation. Companies with more than 20% missing values across all variables were excluded to maintain data integrity and avoid noise from insufficiently populated profiles. Variables with over 50% missing values were also excluded, following standard practice in high-dimensional settings to prevent unreliable imputation from dominating the signal. For the

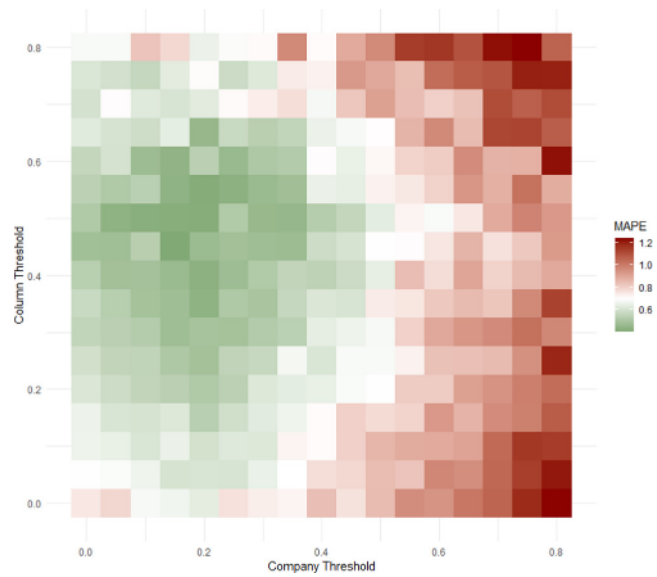

Fig. 2. Sensitivity matrix of the thresholds for eliminating missing values.

remaining missing values, univariate median imputation was applied, which is robust to outliers and well-suited to small datasets, as it preserves central tendency without skewing the distribution. Outliers were identified and capped using a 99.5% winsorisation threshold, ensuring that extreme values did not disproportionately influence model estimates while retaining as much data as possible. Numerical variables were then standardised using z-score transformation when normally distributed or log-transformed for skewed variables, based on Shapiro–Wilk test results and histogram inspection. These preprocessing steps were guided by exploratory data analysis and grounded in best practices for small, high-dimensional datasets [46,47]. Alternative methods, such as k-nearest neighbours (KNN) or multiple imputation, were considered but discarded due to the risk of overfitting and lack of convergence on a small sample with sparse data.

Data augmentation was employed to enhance the training set by introducing random Gaussian noise to selected numerical variables, thereby synthetically generating new observations sampled from the empirical distribution of the original data. This strategy is commonly used to mitigate overfitting in small tabular datasets, particularly when the feature-to-instance ratio is high [48]. For each numerical feature x, synthetic samples $\tilde{x}$ were generated using $\tilde{x} = x + \mathcal{N}(0, \sigma^2 \cdot \gamma^2)$, where σ is the empirical standard deviation of x, and γ is a tunable scaling factor. Categorical features remained unchanged across synthetic records to preserve logical consistency. Since the data augmentation process was applied exclusively to the training set and no synthetic records were

Table 3
Statistical summary of the intermediate valuations dataset after data preparation.

Category	Valuation methodology	Median (millions)	σ (millions)	% NAs
Balance sheet	Book Value	0.34	25.40	3.70%
Income statement	Graham's IV	(1,975.85)	9,817.60	18.52%
Multiples	Benchmark	334.16	8,033.17	62.96%
Discounted cash flows	DCF	16.31	93.19	44.44%
	Terminal Growth	(34.77)	95.26	77.78%
Value creation	EVA	(0.38)	14.01	66.67%
	CVA	16.94	284.90	81.49%
	TCO	14.64	40.55	0.00%
Scorecard	BSC	7.82	201.28	7.40%
	Checklist Method	0.81	14.24	7.40%
Direct intellectual capital	VAIC	60.14	121.60	0.00%
Real valuation		211.02	2171.42	0.00%

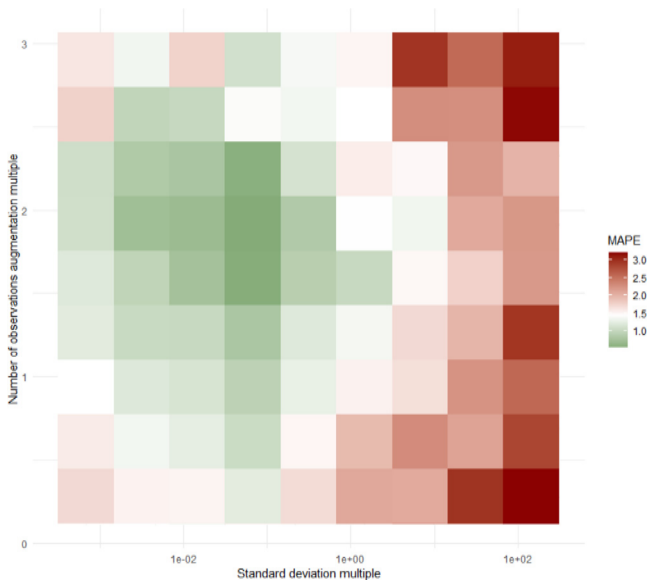

Fig. 3. Sensitivity matrix of the thresholds for data augmentation.

introduced during model evaluation, this technique served solely to improve model stability and mitigate overfitting without compromising the empirical validity of the results.

The number of synthetic observations per original record and the value of γ were selected based on a grid search over a sensitivity matrix (Fig. 3), evaluating model performance using MAPE on a validation set. As illustrated in the figure, optimal performance was achieved with a moderate level of augmentation (2–3 synthetic records per observation) and standard deviation multiples between 10^{-2} and 10^{0}. Extreme values of γ introduced excess noise or overfitting, thereby increasing the error.

While this technique improved model generalisability in cross-validation, we acknowledge that data augmentation can introduce minor distributional shifts. Therefore, results from augmented datasets were always validated against the original to ensure consistency. Future work may explore advanced generative methods, such as variational autoencoders or tabular generative adversarial networks, to further improve sample fidelity [49].

The final dataset consisted of 27 observations and 250 variables after data preprocessing, as shown in Table 4.

3.1.3. Modelling

To evaluate the predictive performance of firm valuation models, we selected a set of ML algorithms that are particularly suited to high-dimensional, low-sample-size settings. These include decision tree-based methods (RF and XGBoost), SVMs, NNs, Ridge regression (RR), KNN, and the C5.0 algorithm. The choice was guided by their proven performance in related finance and entrepreneurial valuation studies (see Table 5), as well as their capacity to handle multicollinearity, sparse predictors, and non-linear relationships.

Simpler models (e.g., OLS, decision stumps) were excluded after preliminary testing due to inadequate performance in terms of MAPE and R^2. Likewise, deep learning approaches, while powerful, were avoided due to the small sample size and the need for interpretability, particularly in VC settings where model transparency is critical.[1]

To further enhance interpretability, LIME was preferred over SHAP due to its localised nature, which is more aligned with the individualised decision-making process in VC. LIME provides case-specific insights by generating locally linear surrogate models that explain individual predictions, which is critical in VC contexts where investment decisions are made on a deal-by-deal basis. However, LIME is known to exhibit instability in high-dimensional, small-sample settings, particularly due to its sensitivity to neighbourhood definition and perturbation sampling. While SHAP offers stronger consistency guarantees and more robust global explanations, its local outputs may be less intuitive for stakeholders unfamiliar with additive feature attribution.

3.1.4. Evaluation

The models were evaluated using two key metrics: R^2 and the MAPE. These metrics were selected for their ability to provide both relative and absolute perspectives on model performance. R^2 quantifies the proportion of variance in the dependent variable that is predictable from the independent variables, while MAPE offers an intuitive measure of forecast error as a percentage, which is particularly interpretable in financial contexts.

To assess the robustness of these metrics in a small-sample, high-dimensional setting, we applied bootstrapping with 1,000 resamples of the training dataset. This procedure enabled the estimation of empirical confidence intervals around both R^2 and MAPE, allowing for an assessment of performance variability, generalisation potential, and statistical significance. The bootstrapped distributions confirmed that the observed performance improvements of the enhanced regression model over the baseline were not due to random variation.

In addition to these accuracy metrics, feature importance was assessed using the Percentage Increase in Mean Squared Error (%IncMSE) and Increase in Node Purity (IncNodePurity). These metrics were derived from permutation-based methods, which measure the impact of each variable on the model's predictive accuracy and structural splits, respectively. Together, these evaluation strategies offer a comprehensive framework to benchmark predictive performance and validate the theoretical constructs embedded in the proposed valuation models.

[1] Available at https://christophm.github.io/interpretable-ml-book/.

Table 4
Company characteristics categories and their representativity in the dataset after data collection and after data preparation.

Categories deducted from the literature	Factors present in the literature	Number of factors in the dataset	
		After data collection	After data preparation
Financial Metrics	Profitability; Revenue growth; Assets under management; Cash-on-cash multiples	228	105
Market and Industry Factors	Market timing; Industry timing; Market potential; Product and technology; Concentration; Market share; Industry structure Differentiation; Industry growth; Google Trends Index industry; Google Trends Index product	53	12
Team and Management	Management team track record; Internal resources; Industry resources; Network-based resources; Strategy; Founder experiences; Founder education; Founder/Technology, media, and telecommunications industrial experience; Management experience; Startup experience; Solo/team dummy; Technology, media, and telecommunications completeness	53	9
Venture Characteristics	Developmental stage; Legal form; Year of founding; Business-to-consumer/Business-to-business; Industry specialisation; Funding round	282	10
Education and Experience	Master of Business Administration degree; Bachelor's degree; Master's degree; Technical education; PhD degree; Managerial experience; Technical experience; Entrepreneurial experience; Consulting experience; Investment banking experience	43	3
Founder and Team Attributes	Entrepreneur's personality; Entrepreneur's experience; Demonstrated leadership in past; Evaluates and reacts to risk well; At least ten times return in 5-10 years; Articulates venture well	17	5
Funding and Financial History	Equity capital before business angel funding; Financial history	18	13
Legal and Regulatory Factors	Legal issues; Initial funding and legal form; Government; Foreign investor	8	5
Market and Competitive Advantage	Size of the market; Sustainability of competitive advantage; Value added of product/service; International scalability	45	5
Investor and Funding Factors	Investors; Capital availability; Motivation; Sensitivity of scenarios; Investor Fit; Business Plan	32	6
Company and Employee Factors	Country; Employees; Current investors; Operations; Product/Service; Business Model; Cash flow rights; Cash flow participation; Cash flow nonparticipation; Business group; Board independence; Blockholder; Multi control; Foreign listing	173	27
Corporate Governance	Board size; Board independence	39	7
Founder and Ownership Structure	Management owners; Pyramids	65	15
Risk and Performance Indicators	Leverage; Capital intensity; Dividend; Age	5	1
Market and Customer Factors	Export; Analyst Following; Market share; Twitter sentiment; Social media presence; No. of tweets; No. of retweets; Negative emotions; Positive emotions; No. of tweet languages; Social Network size	44	9
Product and Service Factors	Value offer; Other assets	90	8
Founder and Team Education	Business education; Uni top business; Uni top tech	6	1
Intellectual Property	Protectability; Proprietary protection	38	5
Investor Perception	Route to market; Relevant experience; Financial model	7	2
Economic Factors	Real interest rate; Consumer price index; Gross domestic product	63	2
Total		**1,308**	**250**

Table 5
Summary of the ML algorithms and their usage in comparable studies.

Algorithm	Studies
Decision trees-based (RF, XGBoost, C5.0)	[20], [23], [38], [50], [51], [52], [53]
SVM	[12], [22], [29]
NN-based	[11], [13], [14], [16], [54]
RR	[21], [24], [55]

3.2. Proposed valuation models

The study proposes four valuation models, each representing different levels of integration between financial metrics, company characteristics, and human decision-making heuristics. These models serve both as predictive instruments and benchmarking tools, enabling comparative evaluation of multiple ML algorithms under real-world constraints.

The first model emulates the manual weight assignment process frequently used in VC firms by combining intermediate valuations with selected company features. It is particularly useful for industry interpretability and was tested with algorithms that align with expert-driven inputs. The second and third models rely solely on company characteristics or combine both input sets to build more generalisable valuation models. The final multilayered integrated model incorporates domain-specific investment restrictions and performs best in environments that emulate real-world VC decision processes.

Each model was tested using seven algorithms and validated using 10-fold cross-validation, a standard technique in small-data settings that enhances robustness by maximising the use of limited observations. In this approach, the dataset was partitioned into ten equally sized, randomly assigned folds without stratification or repetition; each model was trained on nine folds and tested on the remaining one, rotating until all folds served as the test set once. This procedure reduces the risk of overfitting and helps mitigate the influence of outliers on

Table 6
Summary of the frameworks tested by algorithm, inputs, and usage.

Framework	Algorithms tested							Datasets used		Main usage		
	C5.0	KNN	NN	RF	RR	SVM	XGBoost	Company Char.	Intermediate Vals.	Performance Benchmark	Explainability	Industry Process Fit
Regression with intermediate valuations	X	X	X	X	X	X	X		X	X	X	
Regression with company characteristics	X	X	X	X	X	X	X	X		X	X	
Enhanced regression model	X	X	X	X	X	X	X	X	X	X		
Multilayered integrated model with industry restrictions			X					X	X	X		X

Table 7
Regression with Intermediate Valuations framework results.

Algorithm	Median MAPE	σ MAPE	Median R^2	σ R^2
C5.0	**0.456**	1.488	0.810	0.319
XGBoost	0.565	**0.492**	0.946	**0.290**
RF	0.655	1.834	**0.988**	0.298
RR	0.749	6.024	0.876	0.433
KNN	0.835	1.539	0.919	0.326
SVM	0.959	7.025	0.940	0.390
NN	1.730	1.900	0.431	0.411

performance estimation, which is particularly relevant given the small number of observations. Models were evaluated against both industry-standard metrics (MAPE and R^2) and practical criteria, including model transparency, usability, and adoption potential. Table 6 provides an overview of the tested frameworks.

4. Results and discussion

This section presents the findings from the validation of the proposed ML frameworks and analyses the significance of both valuation methodologies and company characteristics. In Section 4.1, the frameworks are validated through performance metrics compared to industry benchmarks. In Section 4.2, the importance of valuation methodologies is assessed, while Section 4.3 focuses on the significance of company characteristics and their correlation with firm value. The broader implications for VC decision-making and entrepreneurial finance are explored in Section 4.4.

4.1. Framework validation

The proposed ML frameworks were validated by comparing multiple algorithms and assessing their predictive accuracy using R^2 and MAPE. The three frameworks tested were: Regression with Intermediate Valuations, Regression with Company Characteristics, and the Enhanced Regression Model.

Table 7 presents the results of the Regression with Intermediate Valuations framework, where XGBoost demonstrates strong predictive accuracy with a balance between low MAPE (0.565) and high R^2 (0.946), although it underperforms relative to the industry benchmark by 13%. Despite this, the results show that ML techniques have the potential to improve valuations in VC by integrating diverse financial metrics.

Table 8 summarises the Regression with Company Characteristics framework results, where XGBoost outperforms other algorithms, reducing the error by 30% compared to industry standards, with a median MAPE of 0.347 and an exceptionally high R^2 of 0.997. This indicates that integrating both financial and non-financial characteristics into ML models leads to significant improvements in predictive accuracy. The Enhanced Regression Model (Table 9) shows similarly strong results, particularly for XGBoost, which surpasses the industry average by 2%.

Table 10 compares the performance across the different frameworks. Notably, the Regression with Company Characteristics framework emerges as the best-performing model, with a 30% improvement over the industry benchmark. This demonstrates that ML models leveraging diverse company characteristics (financial and non-financial) can significantly enhance valuation accuracy in technology-based entrepreneurial settings, offering better tools for investors and venture capitalists.

These results provide important implications for decision-makers in the VC industry. The superior performance of ML models, particularly XGBoost, suggests that venture capitalists can achieve more reliable and accurate valuations by adopting ML-based frameworks that consider both financial and non-financial characteristics. This is particularly relevant for early-stage technology firms, where traditional valuation methods often fail to capture the value of intangible assets like intellectual capital. Implementing these models can help reduce valuation inaccuracies, improve investment decision-making, and potentially reduce risks associated with overvaluations or undervaluations in startup ecosystems.

4.2. Valuation methodology importance analysis

The importance of each valuation methodology was assessed using %IncMSE and IncNodePurity. Table 11 shows that the TCO is the most significant methodology, followed by Book Value and the Checklist Method. These three approaches, grounded in financial statements, account for approximately 63% of the model's predictive accuracy, with TCO displaying a particularly strong %IncMSE of 38.11 (see Fig. 4 for a visual representation).

Interestingly, methodologies incorporating intangible assets, such as VAIC, also contribute significantly (27.09%IncMSE), reflecting the increasing importance of intellectual capital in the valuation of technology-driven ventures. Non-financial valuation methods, such as the BSC and Benchmark, also perform well, suggesting that a hybrid approach combining both financial and non-financial factors provides the most comprehensive valuation model.

Exploratory analyses further suggest that the effectiveness of each valuation methodology is conditional on specific company characteristics. For instance, VAIC exhibited higher local importance for firms with a high research and development intensity, while Book Value showed greater relevance for capital-intensive firms with large tangible

Table 8
Regression with Company Characteristics framework results.

Algorithm	Median MAPE	σ MAPE	Median R^2	σ R^2
XGBoost	**0.347**	**0.950**	0.997	**0.104**
C5.0	0.534	9.628	0.844	0.451
KNN	0.705	3.485	0.993	0.371
NN	0.843	2.671	0.980	0.163
SVM	0.868	2.917	0.949	0.313
RR	0.952	1.911	0.863	0.400
RF	1.130	3.255	**0.999**	0.310

Table 9
Enhanced Regression Model framework results.

Algorithm	Median MAPE	σ MAPE	Median R^2	σ R^2
XGBoost	**0.489**	1.678	**0.996**	0.190
C5.0	0.699	1.632	0.985	0.297
SVM	0.703	3.193	*	*
KNN	0.759	**1.592**	0.943	0.454
RF	0.940	2.252	0.950	**0.109**
NN	1.266	1.709	0.811	0.367
RR	1.277	4.423	0.935	0.392

* Unable to compute.

Table 10
Comparison of the frameworks' results.

Framework	Median MAPE	σ MAPE	Median R^2	σ R^2
Regression with Intermediate Valuations	0.565	0.492	0.946	0.290
Regression with Company Characteristics	**0.347**	0.950	**0.997**	0.104
Enhanced Regression Model	0.489	1.678	0.996	0.190
Multilayer NN w/ Industry Restrictions	0.535	**0.074**	0.983	**0.002**

Table 11
Summary of valuation methodologies importance.

Valuation methodology	%IncMSE	p-value (%IncMSE)	IncNodePurity	p-value (IncNodePurity)
TCO	**38.11**	0.70%	6.63E+17	22.08%
Book Value	31.12	1.40%	**1.01E+18**	5.39%
Checklist Method	29.00	2.20%	6.21E+17	16.78%
VAIC	27.09	2.60%	8.87E+17	9.99%
Graham's IV	26.67	2.70%	3.37E+17	82.02%
DCF	26.20	2.10%	6.31E+17	10.59%
BSC	25.15	5.00%	4.62E+17	62.64%
Benchmark	16.01	3.10%	2.08E+17	24.08%
EVA	6.30	19.68%	5.12E+16	99.30%
Terminal Growth	0.99	26.97%	4.61E+16	99.30%
CVA	−1.15	39.56%	8.26E+16	48.25%

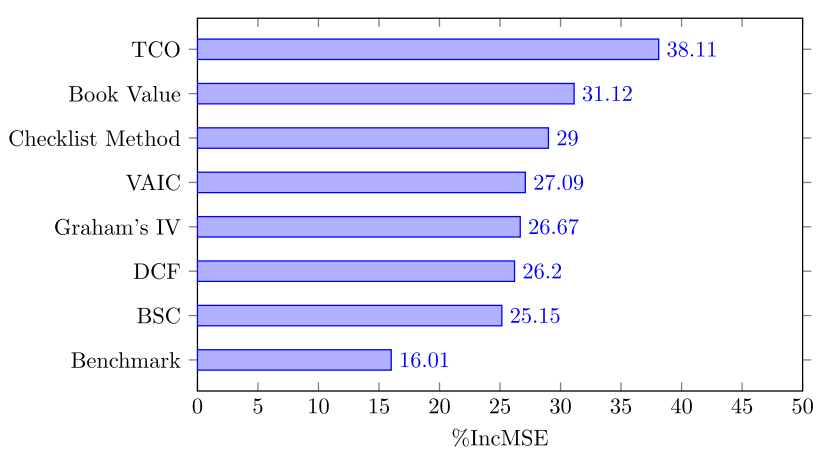

Fig. 4. Variable importance plot of significant valuation methodologies (%IncMSE).

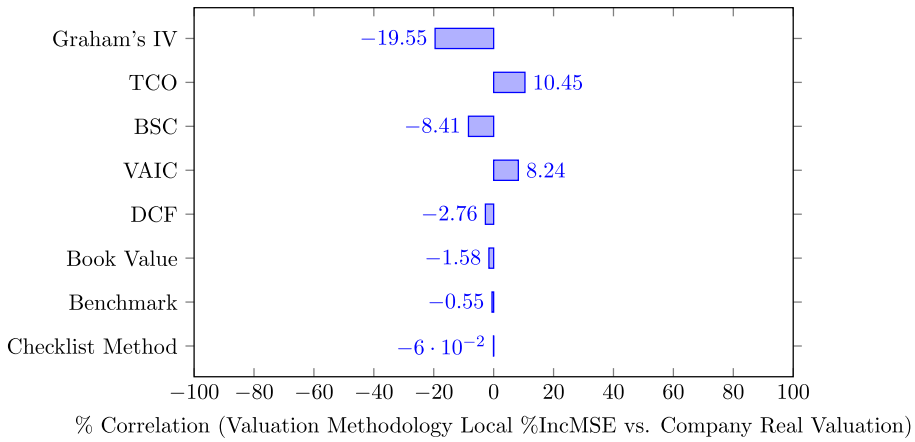

Fig. 5. Correlation between valuation methodologies importance and real valuation.

asset bases. TCO, in contrast, was particularly predictive for early-stage startups with relatively high operational costs. Although formal interaction effects were not modelled in this study, these patterns hint at latent synergies between firm features and intermediate valuation approaches. Future research should explore such interactions using advanced interpretable ML techniques, such as conditional permutation importance or partial dependence diagnostics.

The correlation analysis in Fig. 5 indicates no strong global association between methodology importance and firm valuation. However, TCO and VAIC show a mild positive trend for smaller firms, underscoring their relevance in early-stage assessments, where intangible assets and cost-side considerations dominate. Notably, the statistical significance of these effects is supported by low p-values for %IncMSE (0.70% and 2.60% for TCO and VAIC, respectively), which suggest that these methodologies offer a non-random, substantive contribution to predictive performance. In contrast, high p-values for other methods (e.g., EVA or CVA) indicate a weaker role in explaining valuation outcomes within this dataset.

For investors and venture capitalists, these findings support the adoption of a hybrid valuation strategy. The relative strengths of TCO and VAIC indicate that blending traditional financial data with indicators of intellectual capital and cost structures can improve the predictive validity and transparency of early-stage firm valuations. This integrated approach can inform due diligence, enhance investment decision-making, and reduce risks associated with over- or undervaluation.

4.3. Company characteristics importance analysis

Table 12 provides a summary of the results for the most significant company characteristics. Financial metrics dominate the model, with 37 out of 54 statistically significant factors, contributing the highest cumulative %IncMSE (433.08). This aligns with the literature on firm valuation, where financial performance indicators are critical for assessing company value.

The statistical robustness of these variable importance measures is further underscored by the low p-values associated with %IncMSE and IncNodePurity across the majority of financial factors. For instance, 33 out of 37 financial variables show p-values below 5%, indicating that their predictive contribution is unlikely due to chance. This adds empirical support to the use of financial metrics as a reliable basis for firm valuation. However, some non-financial categories (such as Team and Management) also demonstrate statistically significant influence, highlighting the need to integrate quantitative and qualitative predictors in ML-based valuation models.

However, non-financial factors such as intellectual property, founder experience, and market potential also play a substantial role, particularly in early-stage firms. Table 13 shows that while financial metrics are the dominant category, non-financial characteristics (e.g., team and management, intellectual property) account for approximately 17% of the model's predictive power.

The correlation analysis in Fig. 6 shows that financial metrics, particularly operating expenses and revenues, have a positive correlation with firm value, while non-financial factors like cash reserves and target market size are more significant for smaller companies. Fig. 7 highlights that as firms grow, the importance of these non-financial factors decreases, suggesting that early-stage firms require a different valuation approach than more established companies.

Fig. 7 shows the correlation between the importance of two of the eight factors with the highest correlation with real company valuation.

For venture capitalists and other investors, the findings suggest that financial metrics should remain the primary focus for valuing mature companies. However, in the case of startups and early-stage firms, non-financial factors such as intellectual capital and market potential should be weighted more heavily in valuation models. This dual approach can provide a more accurate reflection of a company's potential, allowing investors to make more informed decisions about where to allocate capital. The results also encourage venture capitalists to adopt ML models that integrate both financial and non-financial data to improve the accuracy of early-stage valuations, leading to better investment outcomes in technology-based firms.

4.4. Entrepreneurial valuation in a data-driven landscape

The findings from the preceding sections highlight the growing complexity of startup valuation, particularly in industries where traditional financial metrics fail to capture the full scope of firm potential. The results in Table 10 confirm that ML models, particularly those incorporating non-financial characteristics, improve predictive accuracy beyond conventional approaches. However, the increasing reliance on computational models raises broader questions about how venture capitalists and entrepreneurs can integrate these advancements into their strategic decision-making.

One of the most striking findings is the extent to which non-financial indicators influence startup valuations. Table 12 demonstrates that while financial performance remains the dominant predictor, factors such as intellectual capital, founder experience, and market positioning contribute significantly to firm valuation. These characteristics are often difficult to quantify using traditional regression-based approaches, underscoring the need for models that adapt dynamically to the realities of early-stage entrepreneurial ventures.

A key implication of these findings is the potential for valuation models to move beyond static assessments and evolve into dynamic,

Table 12
Summary of predictive factors importance.

Category	No. factors	%IncMSE *Sum*	*p*-value *Mean—%IncMSE*	IncNodePurity *Sum*	*p*-value *Mean—IncNodePurity*
Financial Metrics	**37**	**433.08**	1.34%	**4.36E+18**	5.73%
Founder and Ownership Structure	4	18.17	1.92%	1.14E+16	28.30%
Market and Industry Factors	2	14.11	2.45%	2.57E+16	59.84%
Team and Management	2	9.49	4.40%	1.05E+17	10.89%
Corporate Governance	2	5.40	2.55%	7.95E+14	42.41%
Company and Employee Factors	2	4.27	4.50%	1.11E+15	59.69%
Funding and Financial History	2	4.11	4.25%	8.77E+14	54.40%
Risk and Performance Indicators	1	2.73	4.70%	7.21E+15	31.07%
Product and Service Factors	1	2.10	4.20%	2.99E+14	55.34%
Intellectual Property	1	1.90	4.90%	4.20E+13	100.00%

Table 13
Summary of company characteristics categories' significance and representativity.

Category	Number of factors in dataset	Number (%) of significant factors	Average %IncMSE
Financial Metrics	**105**	**37** (35.24%)	**11.70**
Company and Employee Factors	27	2 (7.41%)	2.13
Founder and Ownership Structure	15	4 (26.67%)	4.54
Funding and Financial History	13	2 (15.38%)	2.05
Market and Industry Factors	12	2 (16.67%)	7.06
Market and Customer Factors	9	0 (0.00%)	0.00
Social Factors	9	0 (0.00%)	0.00
Team and Management	9	2 (22.22%)	4.75
Product and Service Factors	8	1 (12.5%)	2.10
Corporate Governance	7	2 (28.57%)	2.70
Financial Factors	6	0 (0.00%)	0.00
Market and Competitive Advantage	5	0 (0.00%)	0.00
Intellectual Property	5	1 (20.00%)	1.90
Founder and Team Attributes	5	0 (0.00%)	0.00
Legal and Regulatory Factors	5	0 (0.00%)	0.00
Venture Characteristics	4	0 (0.00%)	0.00
Investor Perception	2	0 (0.00%)	0.00
Economic Factors	2	0 (0.00%)	0.00
Risk and Performance Indicators	1	1 (100.00%)	2.73
Founder and Team Education	1	0 (0.00%)	0.00

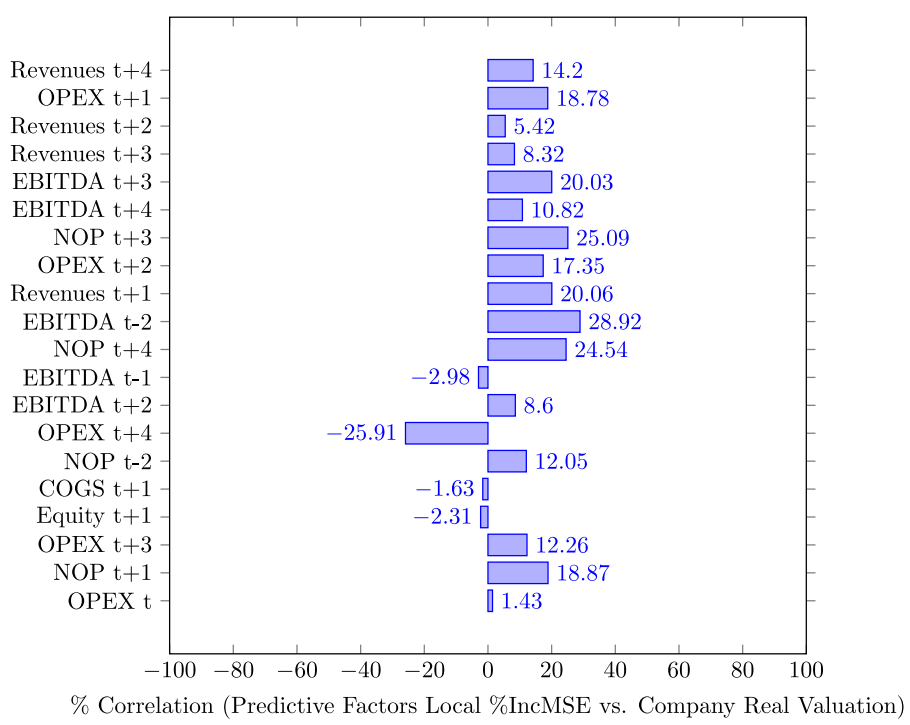

Fig. 6. Correlation between the most important predictive factors' importance and real valuation.

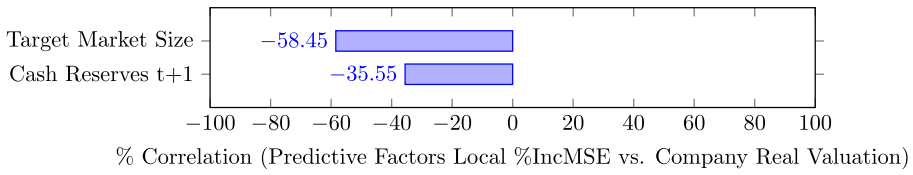

Fig. 7. Correlation between the most correlated predictive factors' importance and real valuation.

continuously updated frameworks that adjust to market conditions in real time. The concept of integrating live operational and financial data streams into predictive models could significantly enhance investors' ability to assess startup potential with greater precision. The strong correlations observed in Fig. 6 between financial performance and valuation provide an opportunity for venture capitalists to simulate future business scenarios, adjusting investment strategies accordingly. In practice, this could involve the development of data-driven valuation ecosystems, where firms' evolving market performance, customer acquisition rates, and operational efficiency inform their real-time valuation rather than relying solely on historical financial statements.

Furthermore, the results suggest that high-performing ML models not only improve valuation accuracy but also provide a foundation for more transparent decision-making. The ability to interpret these models, rather than treating them as opaque "black boxes", is critical for their adoption in investment contexts. The high explanatory power of approaches such as SHAP and LIME, which decompose ML predictions into comprehensible components, enables investors to better understand which factors drive firm value at different stages of growth. This interpretability is particularly important in the context of early-stage startups, where a company's future success often depends on intangible assets such as brand strength, technological capabilities, or founder expertise.

An additional dimension of these findings is the increasing role of computational simulations in VC decision-making. The substantial variation in predictive accuracy across different models (Table 9) suggests that static valuation frameworks struggle to capture the complexity of high-growth startups. A more flexible approach — where valuation models are continuously recalibrated based on emerging data — could mitigate the risk of mispricing investments, particularly in sectors with high levels of uncertainty. Simulated financial and operational projections, built upon the same ML-driven frameworks analysed in this study, would enable venture capitalists to test different strategic scenarios before committing capital. By extending predictive modelling beyond valuation itself, investors can explore how various funding structures, competitive responses, and external market shocks could impact startup performance over time.

Beyond improving individual investment decisions, the broader adoption of AI-driven valuation models could influence the entrepreneurial finance landscape at a systemic level. As seen in Fig. 7, early-stage valuation discrepancies often stem from over-reliance on single financial indicators, leading to inflated expectations or underestimated risks. The introduction of more sophisticated, adaptive valuation mechanisms has the potential to reduce volatility in startup funding markets, improving capital allocation efficiency across different sectors. This shift could have significant implications for the way investors structure their portfolios, prioritising firms that demonstrate sustainable long-term value rather than short-term speculative gains.

Taken together, these findings reinforce the idea that modern entrepreneurial valuation is no longer confined to static financial ratios but is increasingly shaped by advanced analytical techniques. As startups continue to operate in environments characterised by rapid technological change and evolving business models, the ability to integrate real-time data into valuation frameworks will be essential for investors and entrepreneurs alike. The results of this study provide a compelling case for rethinking how early-stage companies are assessed, highlighting the need for more adaptive, explainable, and data-driven approaches to startup valuation.

5. Conclusion

This study developed and validated an advanced ML framework to improve startup valuation processes in the VC industry. The proposed models demonstrated a substantial 30% improvement in valuation accuracy over industry benchmarks, evidenced by consistently high R^2 values and low MAPE scores. This improvement signifies the potential of ML to address the complexities inherent in valuing early-stage companies.

The research provides key insights into the dynamic interaction between various company characteristics and valuation methodologies. By integrating both traditional financial metrics and non-traditional, intangible factors such as intellectual capital, the study underscores the importance of a hybrid approach to valuation, which offers a more comprehensive understanding of a firm's value. The results suggest that methodologies like TOC and VAIC are particularly valuable for assessing technology-intensive startups, where intangible assets often drive much of the value.

One of the core contributions of this research is the introduction of an industry-adapted framework that simplifies the integration of ML into existing valuation processes. By aligning with real-world valuation practices, the framework reduces adoption resistance within the VC industry and demonstrates how ML techniques can complement traditional approaches. This study's data-driven insights into company characteristics further illustrate how firm value evolves over time, offering practical guidance to investors and stakeholders on optimising their valuation processes.

While the enhanced regression model provided robust performance, challenges related to high collinearity and dimensionality in combining intermediate valuations and company characteristics were noted. These challenges highlight the need for future refinements in ML-based valuation models to address the interaction between financial and non-financial data.

This research contributes to the growing body of literature on ML developments in finance, specifically in the context of firm valuation. The hybrid model proposed here bridges a critical gap in the valuation of early-stage firms by integrating intangible assets, thus advancing the theoretical understanding of firm value creation beyond traditional financial metrics. By positioning ML as a solution to complex valuation problems, the study offers a significant theoretical contribution, demonstrating how emerging data-driven techniques can enhance valuation accuracy and provide deeper insights into startup ecosystems.

For venture capitalists and decision-makers, the findings offer practical applications. The results highlight the importance of adopting ML models to complement traditional valuation techniques, especially when assessing early-stage technology-driven firms. Integrating non-financial factors like intellectual capital and market potential into valuation models can provide investors with a more nuanced understanding of firm value, helping them make more informed investment decisions. This hybrid approach can reduce the risk of overvaluation or undervaluation, improve capital allocation, and ultimately drive better investment outcomes.

Despite its contributions, this study has some limitations. First, the analysis was conducted on a relatively small dataset, with a focus on startups primarily from the Nordic region. This geographic concentration may limit the reliability and external validity of the findings, particularly when applied to startups from other regions or

under different market conditions. Furthermore, the small sample size (although partially mitigated by augmentation) raises the possibility of overfitting, especially in high-dimensional models. As such, the robustness and generalisability of the proposed framework should be further tested using larger, more diverse datasets in future research. The exclusion of certain company characteristics due to data sparsity and privacy concerns further constrained the analysis. Additionally, the reliance on secondary data sources, without complementary primary data collection methods such as structured surveys, interviews, or expert consultations, limited the depth of the analysis and may have overlooked qualitative factors critical to valuation accuracy. This is particularly relevant for startup ecosystems, where founder intent, organisational dynamics, or market narrative often play a significant role in shaping firm trajectories. For example, challenges related to internal strategic alignment (such as those influencing human resource practices in early-stage firms) can materially affect outcomes in ways not captured by quantitative indicators alone [56]. Incorporating such primary data in future research would enrich the explanatory power and interpretability of machine learning valuation models.

Future studies should aim to address these limitations by expanding the dataset to include a more diverse sample of firms, particularly from emerging markets, to enhance the applicability and unreliability of the findings. Incorporating primary data collection methods, such as surveys or interviews, could provide more granular insights into firm-specific valuation drivers that are not captured in secondary data. Moreover, future research should explore the application of deep learning techniques to the proposed framework, investigating how these advanced models could further improve valuation accuracy, particularly for early-stage technology-based enterprises. By continuously refining the proposed framework and testing it in diverse contexts, future research can facilitate the broader adoption of ML models in the finance industry, ultimately leading to more accurate, reliable, and scalable valuation methodologies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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